

1. Project Overview

The **Hospital Information Management System (HIMS)** is a comprehensive database solution designed to manage hospital operations such as patient records, doctor management, appointments, billing.

This project demonstrates **complete SQL implementation** from **basic queries to advanced concepts**, simulating real-world hospital workflows.

2. Objectives

- Maintain patient and doctor records efficiently
- Automate appointment scheduling
- Manage billing and payments

3. System Modules

Patient Management

- Register new patients
- Maintain patient history

Doctor Management

- Store doctor details (specialization, availability)
- Assign doctors to patients

Appointment Management

- Book, update, cancel appointments
- Track appointment status

Billing System

- Generate bills for treatments
- Manage payments

Reporting System

- Daily patient visits
- Revenue reports
- Doctor performance

4. Database Design (Tables)

Patients

- patient_id (PK)
- name
- age
- gender
- phone
- address

Doctors

- doctor_id (PK)
- name
- specialization
- phone

Appointments

- appointment_id (PK)
- patient_id (FK)
- doctor_id (FK)
- appointment_date
- status

Billing

- bill_id (PK)
 - patient_id (FK)
 - amount
 - bill_date
-

5. Relationships

- One patient → Many appointments
 - One doctor → Many appointments
 - One patient → Many treatments
 - One treatment → Many prescriptions
 - One medicine → Many prescriptions
 - One patient → Many bills
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6. SQL Concepts Covered

Basic SQL Operations

- Insert patient, doctor, appointment records
- Update and delete records
- Filter using WHERE

Joins

- INNER JOIN → Patient + Appointment
- LEFT JOIN → Doctors with no appointments

Aggregate Functions

- COUNT() → Number of patients
- SUM() → Total revenue
- AVG() → Average billing

Subqueries

- Patients with highest bill
- Doctors with maximum appointments

Window Functions

- RANK() doctors by number of patients
- ROW_NUMBER() for appointments
- **Views**
- Create patient summary view
- Billing report view

Indexes

- Index on patient_id, doctor_id for faster search

Constraints

- PRIMARY KEY
- FOREIGN KEY
- NOT NULL
- UNIQUE

Stored Procedures

- Book appointment
- Generate bill

Transactions

- COMMIT and ROLLBACK for billing

CTE (Common Table Expressions)

- Complex reporting queries

7. Workflow

1. Register patient
2. Add doctor details
3. Book appointment
4. Generate bill

8. Sample Queries

Total Revenue

```
SELECT SUM(amount) AS total_revenue FROM billing;
```

Top Doctor by Appointments

```
SELECT doctor_id, COUNT(*) AS total_appointments  
FROM appointments  
GROUP BY doctor_id  
ORDER BY total_appointments DESC  
LIMIT 1;
```

9. Advanced Features

- Online appointment booking
- Emergency patient handling
- Insurance integration
- Multi-branch hospital management
- Role-based access (Admin, Doctor, Staff)

10. Testing & Validation

- Check foreign key constraints
- Validate billing calculations
- Test trigger execution
- Ensure no duplicate records

11. Security Considerations

- Restrict access using roles
- Encrypt sensitive data
- Backup database regularly

12. Conclusion

The **Hospital Information Management System** demonstrates:

- Complete database design from basic to advanced
- Real-world SQL implementation
- Advanced querying techniques
- Data integrity and automation.